THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The sum of the dates in a 2×2 calendar block is 44. The smallest date in the block is: 1

(a) 8 **(b)** 7 **(c)** 15 **(d)** 6

2. The length of a rectangle is increased by 20% and its breadth decreased by 10%. If the original area is 1000 cm², the new area (in cm²) is: 1

(a) 1200 **(b)** 1000 **(c)** 1100 **(d)** 1080

3. The difference of the squares of two consecutive natural numbers is 59. The numbers are: 1

(a) 29 and 31 **(b)** 28 and 29 **(c)** 30 and 31 **(d)** 29 and 30

4. The ones digit of 7^{23} is: 1

(a) 7 **(b)** 9 **(c)** 3 **(d)** 5

5. The number of 1s in the binary form of 77 is: 1

(a) 6 **(b)** 4 **(c)** 3 **(d)** 5

6. The diagonals of a rectangle ABCD meet at O. If $\angle AOD = 100^\circ$, then $\angle OAB$ is: 1

(a) 50° **(b)** 65° **(c)** 100° **(d)** 40°

7. The smallest 3-digit palindrome divisible by 3 is: 1

(a) 111 **(b)** 101 **(c)** 114 **(d)** 121

8. If $x + \dfrac{1}{x} = 3$, then $x^2 + \dfrac{1}{x^2}$ equals: 1

(a) 11 **(b)** 9 **(c)** 8 **(d)** 7

9. 12 men can finish a job in 20 days. After 8 days, 4 more men join. The remaining work will be finished in: 1

(a) 12 days **(b)** 9 days **(c)** 11 days **(d)** 15 days

10. A price is first increased by 20% and then decreased by 20%. The net change (as percent, + for increase, − for decrease) is: 1

(a) −3% **(b)** −4% **(c)** 0% **(d)** −5%

11. A rhombus has perimeter 292 cm and one diagonal 96 cm. Its area (in cm²) is: 1

(a) 5328 **(b)** 5280 **(c)** 7008 **(d)** 2640

12. If $a : b = 3 : 5$ and $b : c = 10 : 7$ and $a + b + c = 322$, then b equals: 1

(a) 84 **(b)** 140 **(c)** 154 **(d)** 98

13. A cube is painted all over and cut into unit cubes. If exactly 24 of the unit cubes have exactly one face painted, the side of the big cube is: 1

(a) 6 **(b)** 4 **(c)** 5 **(d)** 3

14. The mean of the first 12 odd natural numbers is: 1

(a) 13 **(b)** 6.5 **(c)** 144 **(d)** 12

15. If $a^2 - b^2 = 144$ and $a - b = 8$, then $a + b$ equals: 1

(a) 65 **(b)** 8 **(c)** 19 **(d)** 18

16. The area of a triangle is 120 cm². If the altitude on the side of 20 cm is 12 cm, the altitude on the side of 16 cm (in cm) is: 1

(a) 7 **(b)** 12 **(c)** 17 **(d)** 15

17. **Assertion (A):** A triangle with sides 6 cm, 8 cm and 10 cm is right-angled. 1
Reason (R): If the square of the longest side equals the sum of the squares of the other two sides, the triangle is right-angled.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** If 5 kg of rice costs ₹300, then 8 kg of rice costs ₹480. 1
Reason (R): The cost of rice is directly proportional to its quantity, so $\frac{\text{cost}}{\text{quantity}}$ is constant.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** A cube satisfies Euler's formula. 1
Reason (R): For every convex polyhedron, $F + V - E = 2$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** The mean of 4, 6, 8, 10 and 12 is 8. 1
Reason (R): The mean of a data set is the sum of the observations divided by the number of observations.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. The sum of the digits of a two-digit number is 9 and the tens digit exceeds the ones digit by 3. Find the number. 2

22. The perimeter of a rectangle is 100 cm and its length exceeds its breadth by 18 cm. Find its area. 2

23. The difference between the squares of two consecutive numbers is 75. Find the numbers. 2

24. Write $(3 \times 10^5) + (4 \times 10^4)$ in scientific notation. 2

25. Write 90 in binary and count the number of 1s in it. 2

26. The angles of a quadrilateral are in the ratio 2:5:6:7. Find the largest angle and say whether it is acute, right or obtuse. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Add 343_5 and 212_5 (numbers in base 5) and give the answer in base 5. 3

28. The angles of a quadrilateral are x°, $(2x - 10)^\circ$, $(3x - 20)^\circ$ and $(4x - 30)^\circ$. Find all four angles. 3

29. Find the greatest 4-digit number divisible by 6, 8 and 12. 3

30. Simplify $(3x + 2)^2 - (3x - 2)^2$. 3

31. 15 men can finish a job in 24 days. After 8 days, 5 more men join them. In how many more days is the job finished? 3

OR

The value of a car depreciates by 10% every year. If its present value is ₹ 500000, find its value after 3 years. 3

32. A rhombus has perimeter 292 cm and one diagonal 96 cm. Find the other diagonal and the area. 3

OR

A mixture contains milk and water in the ratio 3:2. When 10 L of water is added, the ratio of milk to water becomes 3:4. Find the quantity of milk and water in the original mixture. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** A school gives each student a four-digit locker code. Neha's code is 7,236 and Tarun's code is 52**d**4, where d is a missing digit. 5

(i) Is 7,236 divisible by 9?

(ii) Is 7,236 divisible by 4?

(iii) Is 7,236 divisible by 11?

(iv) Find the digit d so that 52d4 is divisible by 9, and check that the number is then also divisible by 6.

34. **Case study.** A rectangular vegetable plot has length $(x + 5)$ m and breadth $(x + 2)$ m. A square flower bed beside it has side $(x + 3)$ m. 5

(i) Write the area of the vegetable plot as an expression.

(ii) Find its area when $x = 10$.

(iii) Write the perimeter of the plot.

(iv) By how much does the area of the plot exceed that of the flower bed? Verify for $x = 10$.

35. A vessel holds 60 L of a mixture of milk and water in the ratio 2 : 1. 15 L of the mixture is removed and replaced by 15 L of water. Find the new ratio of milk to water. 5

36. A shopkeeper marks an article 50% above its cost price. During a sale he gives a discount of 20% and still earns a profit of ₹ 240. Find the cost price. 5

OR

A ladder 65 m long reaches a point 60 m up a wall. Its top slides down 4 m. Find 5

(i) the original distance of the foot from the wall,

(ii) its new distance from the wall,

(iii) how far the foot moves.

37. 24 workers can finish a job in 30 days. After 10 days 8 workers leave. In how many total days will the job be finished? By how many days is it delayed? 5

OR

A Koch snowflake is built from an equilateral triangle of side 27 cm. Find its perimeter at stage 3, and by how much the perimeter has increased from stage 0. 5

38. The mean of 12 observations is 15. If each observation is multiplied by 4 and then decreased by 5, find the new mean and the new sum. 5

OR

A three-digit number has digits a, b, c with $a - c = 1$. Show that the number minus its reverse is divisible by 99 and find the difference. 5

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND